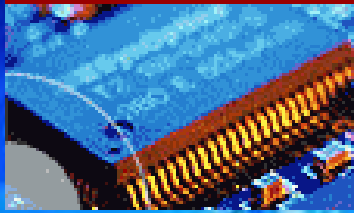


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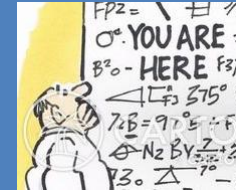
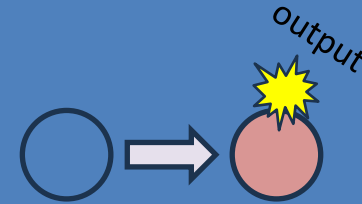
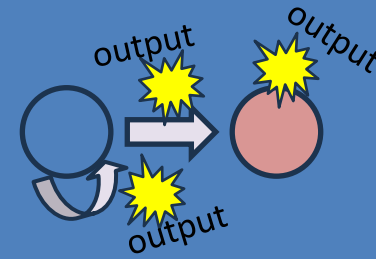
**Embedded
Systems II**

Terms & Concepts

Mealy
vs.
Moore
vs.
Kripke



$$\curvearrowright \subseteq W \times W$$



Finite State Machines FSM

Lecture L12

Embedded Systems II

Prof. Simon Winberg

Should really be lecture 3



Electrical Engineering
University of Cape Town

Outline

- Discussion – who knows FSMs? What did you do with them?
- What is a FSM

Modeling Dynamic Behaviour

- Consider **use cases** to capture:
 - RT and QOS requirements
 - Define fundamental operations
 - Determine communication needs
- Dynamic behaviour is a refinement of these.
 - Defines the **operation** and **behaviour** of classes

Finite State Machines (FSM)

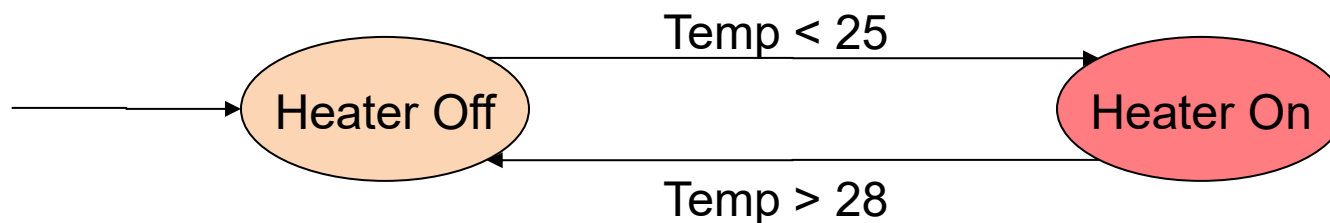
- Using FSMs, the system is modeled in terms of
 - States
 - Transitions between states
- FSMs have several advantages for system modeling
 - The approach is visual and intuitive
 - The approach is mathematically rigorous if approached carefully
 - Is applicable to many types of systems

Finite State Machines (FSM)

- FSM are highly *abstracted* models
- Many real-world details are left out:
 - How inputs are obtained
 - How often inputs are checked
 - How states are represented
- Due to this, they greatly simplify systems
 - This helps in analysis and debugging

Different Approaches to FSM

- Several different approaches to FSM exist
 - All provide similar features
 - Three types are widely used:
 - Mealy (edge & state labeled) machines
output depends on current state & input
 - Moore (edge labeled) machines
output depends on only current state
 - Kripke (state labeled) structures

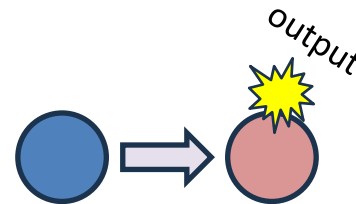
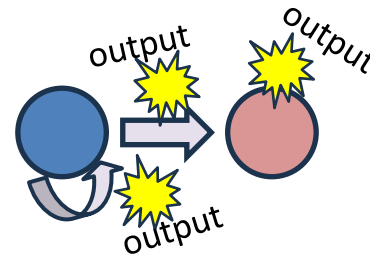


Moore FSM Model of Thermostat Controlled Heater

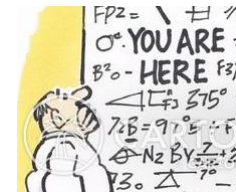
Statemachines

- A state machine has:
 - Input events
 - Output events
 - Set of states
 - A function that maps
 $(\text{state}, \text{input}) \rightarrow (\text{state}, \text{output})$
 - A indication of the initial state
- A Finite State Machine (FSM) has a limited number of states

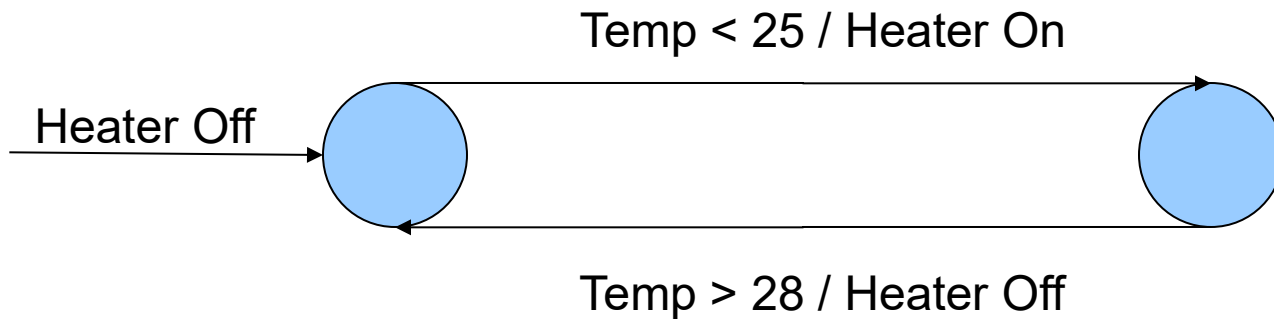
Mealy
vs.
Moore
vs.
Kripke



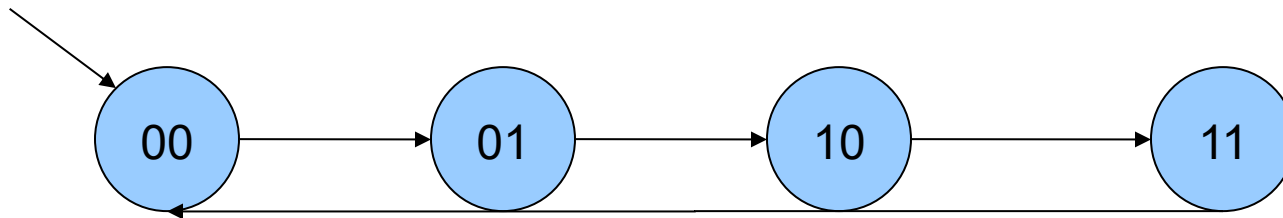
$$\rightarrow \subseteq W \times W$$



Different Approaches to FSM



Mealy FSM Model of Thermostat Controlled Heater



Kripke Structure Model Of Thermostat Controlled Heater

Case Study

Simple scenario for developing and testing a state machine ...

(some stories and reflection to maybe start or end the case study section with)

Determinism In FSM

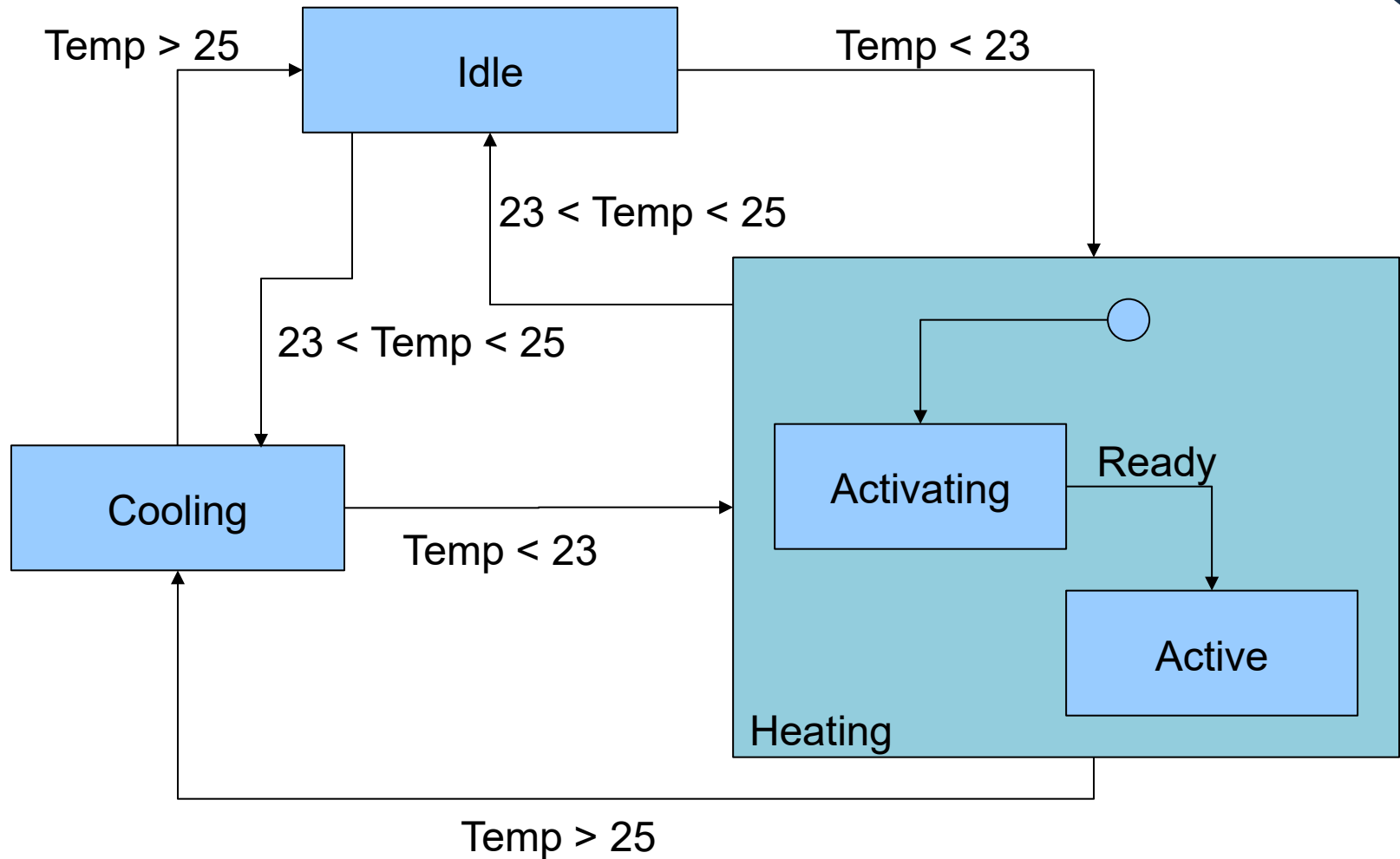
- FSMs are broken up into two categories:
 - Deterministic FSM (D-FSM)
 - For each pair of input and state there is one possible next state
 - Non-Deterministic FSM (ND-FSM)
 - For each pair of input and state there is one or more possible end states
- Both models are mathematically equivalent, but an ND machine could require an extremely complex deterministic representation

Nondeterminism In FSM

- Non determinism often comes into FSM models through **abstraction**
 - The input needed to choose a next state is left out of the model
 - This is often because the model is focusing on another aspect of the system
- Nondeterminism can be difficult to translate into the real world
 - A separate model might resolve the problem

The UML Approach to FSM

Optional Reading



Generally uses a Mealy machine

Adapted From "The UML User Guide", Booch et. al.

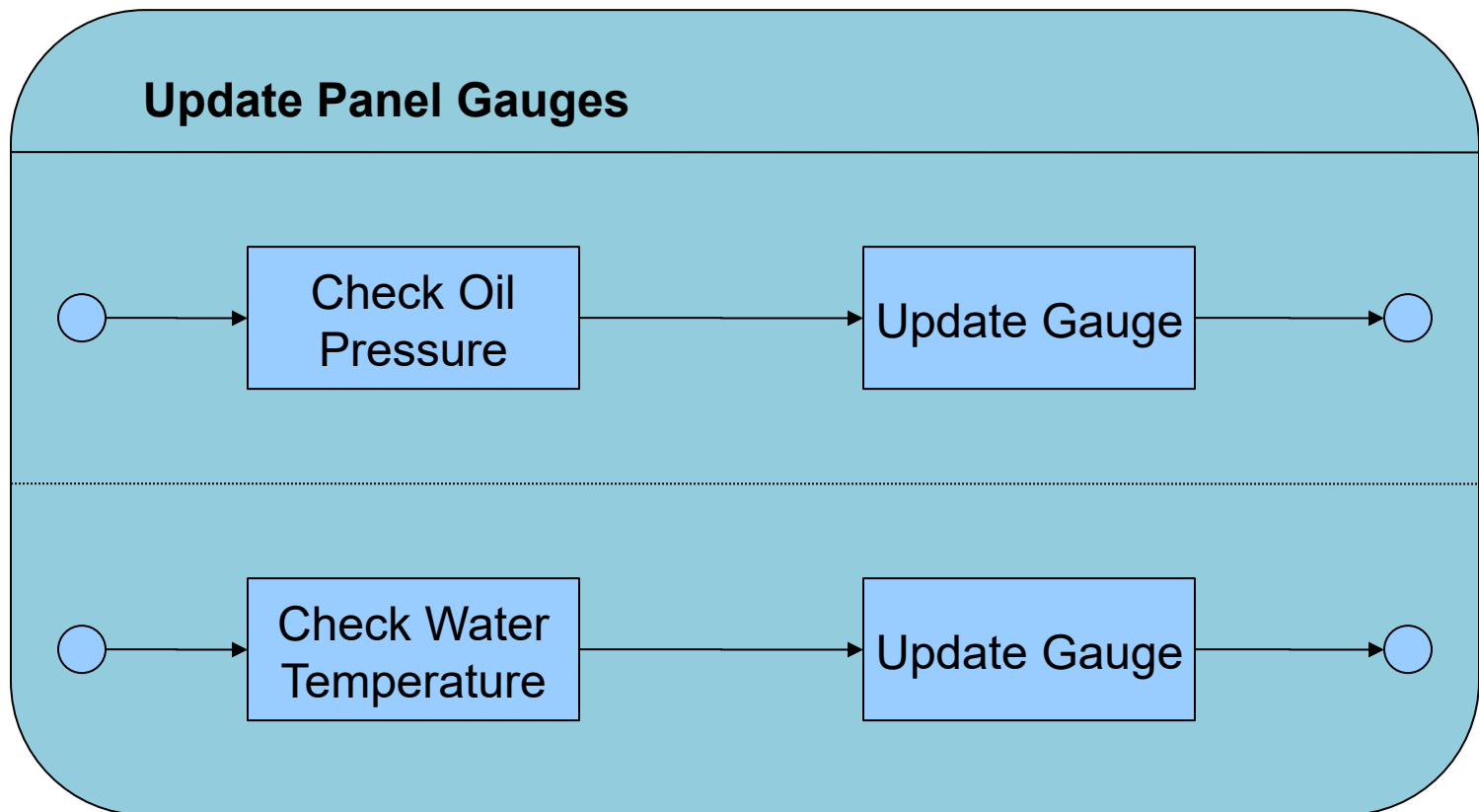
The Shortcomings of FSM

- FSM is ideal for simple systems with 10s of states
 - This category covers many real systems
- Hierarchical FSMs more the way for real systems
- FSM models are often too complex for big systems with 100s or 1000s of states
- Can be improved (for our needs) with:
 - Better, more expressive state structures
 - Methods of combining multiple models

Showing Concurrency

The example below is how concurrency can be shown in a statemachine.

This example is actually of a UML Statechart; don't you have to use that standard.



State Machine Activity

Class Activity

An environmental test chamber is used to test how a smoke detector product performs under certain environmental conditions. A typical test regime consists of increasing the chamber's internal temperature to 75 degrees Celsius. The temperature is maintained by toggling between the chamber's heat-up and cool-down states. This internal temperature is maintained for the test duration of 3 hours. Once the test time elapses, the chamber enters its cool-down state and lowers its internal temperature to room temperature (say 21 degrees Celsius). While the chamber monitors its internal temperature and the test duration timer, it also monitors the device being tested for a fault signal by polling it every 2 seconds. If the device under test generates a fault signal, the test is immediately ended and the chamber enters the idle state.

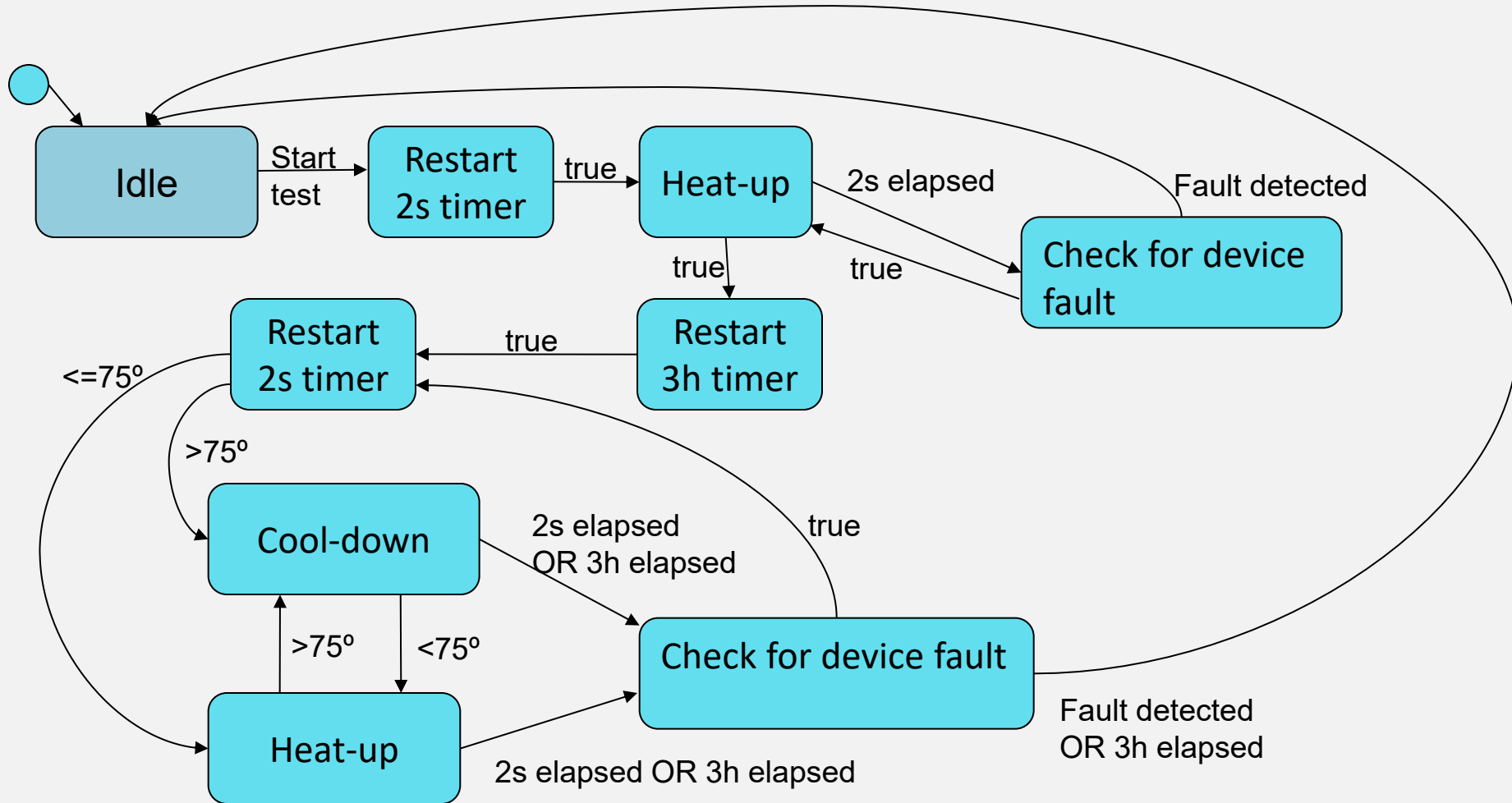


Task:

Draw a state machine representing the typical device test cycle described above. Represent the end of a test cycle as the 'idle' state.

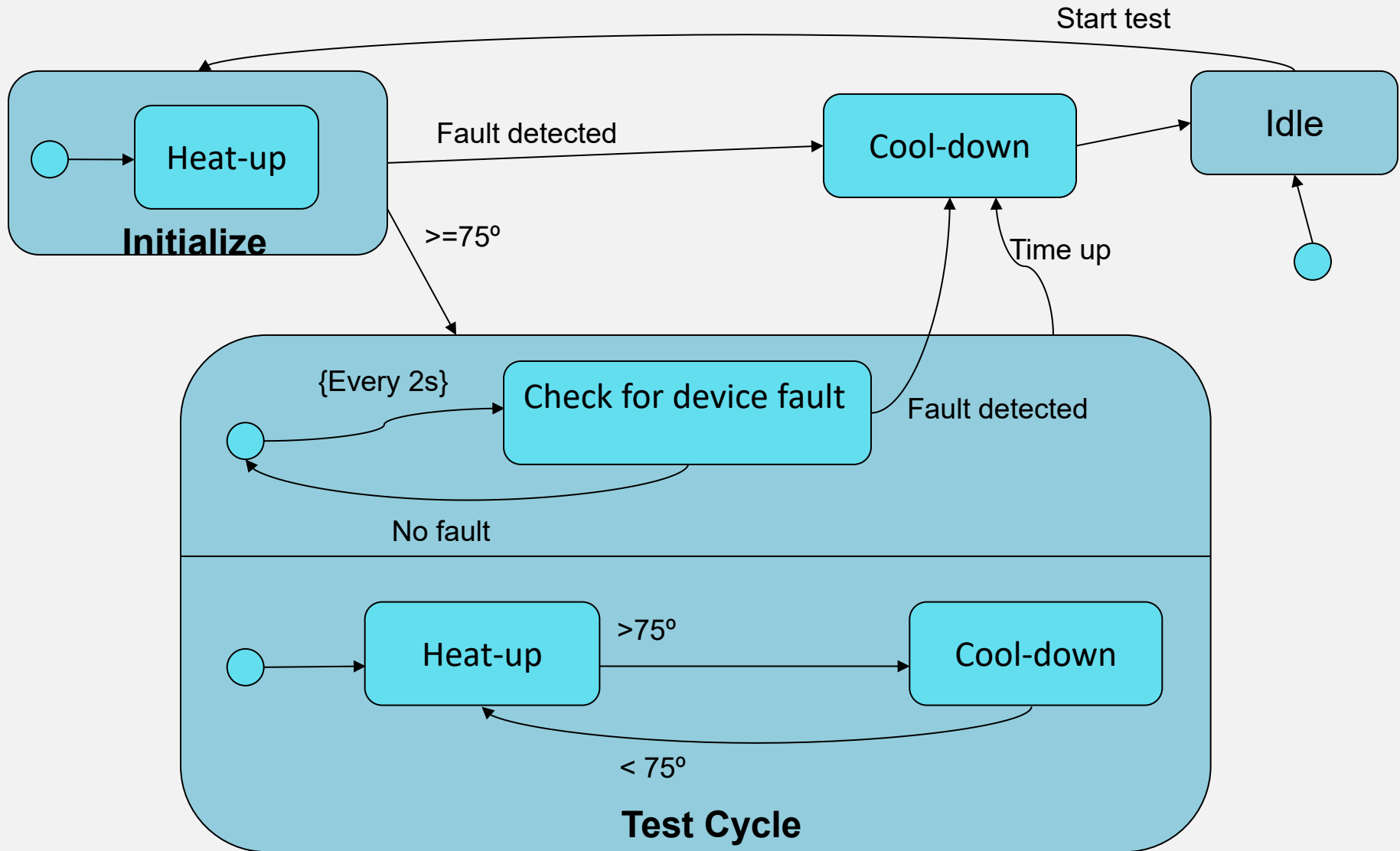
State Machine Sample Solution

See next slide for example that uses a concurrent super-state

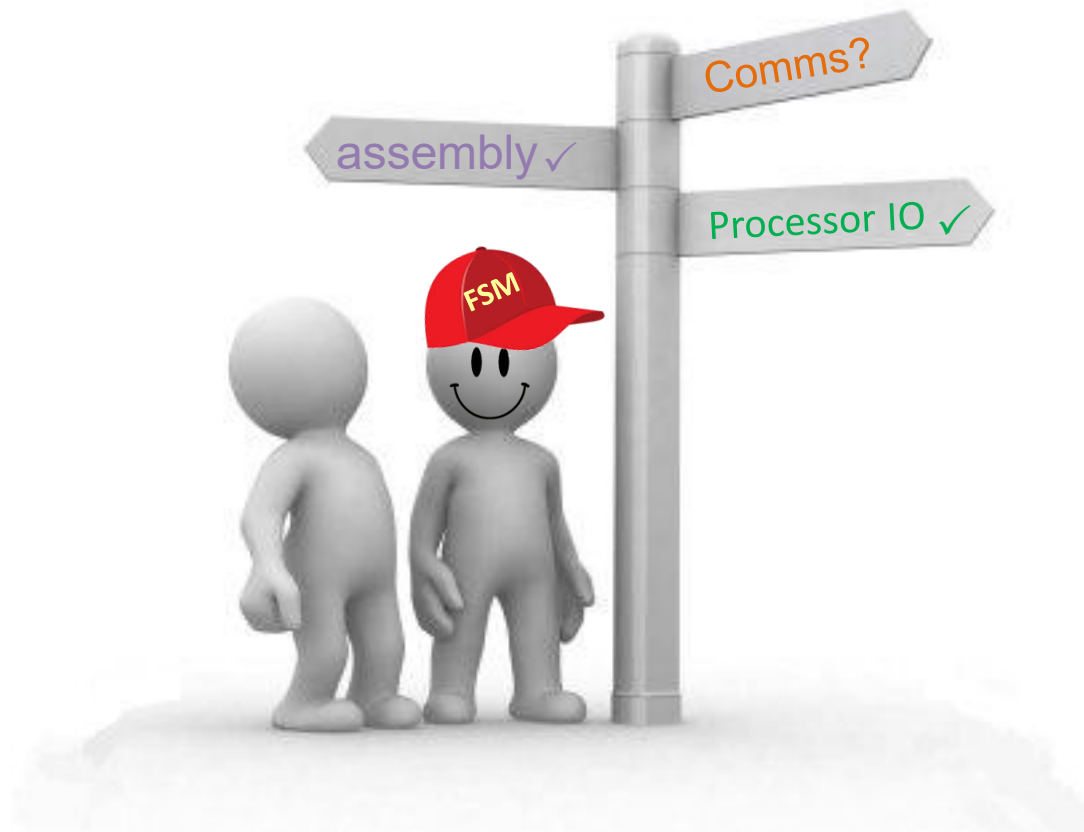


*Using a Moore machine
(outputs depends on current state not in transitions)*

State Chart Sample Solution



FSM & Statecharts



The Next Episode...

Lecture L13

Next Lecture:
Start of Embedded Communications